$$\Rightarrow \qquad CP^2 \text{ is } \begin{cases} > r^2 & \text{or,} \\ = r^2 & \text{or} \quad \{CQ = r\}, \\ < r^2. \end{cases}$$

$$\Rightarrow \quad (x_1+g)^2+(y_1+f)^2 \text{ is } \begin{cases} > g^2+f^2-c & \text{or,} \\ = g^2+f^2-c & \text{or,} \\ < g^2+f^2-c. \end{cases}$$

$$\Rightarrow \quad x_1^2+y_1^2+2gx_1+2fy_1+c \text{ is } \begin{cases} > 0 & \text{or,} \\ = 0 & \text{or,} \\ < 0. \end{cases}$$

Example 5.5

Examine the position of the point $(2,3)$ with respect to the circle $x^2+y^2-6x-8y+12=0$.

Solution

Taking (x_1, y_1) as (2, 3), we get

$$x_1^2+y_1^2+2gx_1+2fy_1+c = 2^2+3^2-6\times 2-8\times 3+12,$$
$$= 4+9-12-24+12 = -11<0.$$

Therefore, the point $(2,3)$ lies inside the circle, by theorem 5.3.

Example 5.6

The line $3x+4y-12=0$ meets the coordinate axes at A and B. Find the equation of the circle drawn on AB as diameter.

Solution

Writing the line $3x+4y=12$, in intercept form yields $\frac{x}{4}+\frac{y}{3}=1$. Hence the points A and B are $(4,0)$ and $(0,3)$.

Equation of the circle in diameter form is

$$(x-x_1)(x-x_2)+(y-y_1)(y-y_2) = 0$$
$$(x-4)(x-0)+(y-0)(y-3) = 0$$
$$x^2+y^2-4x-3y = 0.$$

Example 5.7

A line $3x+4y+10=0$ cuts a chord of length 6 units on a circle with centre of the circle $(2,1)$. Find the equation of the circle in general form.

Solution

$C(2,1)$ is the centre and $3x+4y+10=0$ cuts a chord AB on the circle. Let M be the midpoint of AB. Then we have

$AM = BM = 3$. Now BMC is a right triangle.

So, we have $CM = \dfrac{|3(2)+4(1)+10|}{\sqrt{3^2+4^2}} = 4.$

By Pythogoras theorem $BC^2 = BM^2+MC^2 = 3^2+4^2 = 25$.

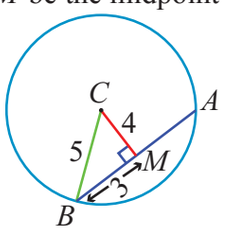


Fig.5.11

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